GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 50° angle. A scale copy is made at k = 2. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 25°
- ● **C. 50°**
- ○ D. 52°

C

Lengths scale; angles do not.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². Find k, and explain why k is not 16. *1 pt*

k = 4. The area ratio is k², because area is two lengths each multiplied by k; 16 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 4.47 m^2 (1.83 m by 2.44 m)

6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft^2 × 0.305^2 = 4.4652. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 11 in? *1 pt*

27.94 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 90 cm? *1 pt*

35.1 in (35.4 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 5 yd? *1 pt*

457.2 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 9 gal? *1 pt*

34.065 L

more litres than gallons

Scoring: 1 point.

14. 1 ft = 12 in. A tile is 7 ft^2 in area. How many square inches is that? *1 pt*

- ● **A. 1,008 in^2**
- ○ B. 84 in^2
- ○ C. 144 in^2
- ○ D. 168 in^2

A

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It starts a 1,500-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second colour. How many gallons is the whole batch? *1 pt*

32 gal

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 450-km drive. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how long? Explain. *1 pt*

Truck B: 55 mi/h = 88.55 km/h < 90 km/h, so 5 h against about 5.08 h — about 5 minutes sooner.

The point needs the speeds in one unit, the choice, and the times (or the gap). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.